

Seasonal Employment: Replication and Extension

Coglianesse & Price (2025) + QWI 6-Digit NAICS Index

July 11, 2026

Outline

- 1 Replication of Coglianese & Price (2025)
- 2 QWI 6-Digit NAICS Seasonal Index
- 3 Has Seasonality Changed Over Time?
- 4 Geographic Variation
- 5 County-Level Exposure (California Pilot)
- 6 Flow vs. Stock: Two Views of Seasonality
- 7 Income/Earnings Seasonality
- 8 Discussion

Coglianese & Price (2025): Overview

- “Income in the Off-Season: Household Adaptation to Yearly Work Interruptions,” *ILRR* 2025
- **Question:** How do households smooth income when employment separations recur annually (i.e. are *seasonal*)?
- **Key innovation:** Identify seasonal separations from administrative data without requiring industry or occupation labels
- Core estimand: **excess recurrence** at 12-month horizon

$$\widehat{\text{ExR}} = \text{Pr}(\text{sep at } k=12) - \frac{\text{Pr}(\text{sep at } k=11) + \text{Pr}(\text{sep at } k=13)}{2}$$

- Counterfactual = average of adjacent lags $\rightarrow$ isolates the *annual spike* marking genuine seasonal separations

Event-study OLS on separators panel (CPS and SIPP):

$$\mathbf{1}[\text{sep at } k] = \sum_j \beta_j \mathbf{1}[k = j] + \varepsilon$$

- Estimate by OLS; excess recurrence = $\beta_{12} - (\beta_{11} + \beta_{13})/2$
- Standard error via delta method
- **Two specifications:**
 - ① *Raw* (Figure 2 in paper): only horizon dummies, no constant
 - ② *Preferred* (Appendix F.1): adds controls for interview-seam artifacts (SIPP) and weeks elapsed since separation (CPS)
- Python replication uses `statsmodels` WLS with clustered SEs; data from the LEHD-style pre-cleaned `.dta.gz` files in the replication package

Replication Results

Table: Excess recurrence estimates: paper vs. Python replication

Specification	Where	Paper	Replication
CPS raw (Figure 2)	Figure 2	1.4 p.p. (SE 0.14)	1.44 ✓
CPS preferred	App. F.1	1.5 p.p. (SE 0.14)	1.42 ✓
SIPP raw (Figure 2)	Figure 2	2.0 p.p. (SE 0.11)	2.04 ✓
SIPP preferred	App. F.1	1.6 p.p. (SE 0.11)	1.62 ✓

- All four estimates within rounding of published values
- Industry ranking (both datasets): Agriculture > Entertainment > Education > Construction ≫ Healthcare, FIRE

Extending to 6-Digit NAICS via QWI

- CP's classifier operates at the worker level; industry rankings are reported at 1-digit NAICS only
- **Goal:** produce a finer, industry-level seasonal index suitable for firm classification
- **Data:** Census Quarterly Workforce Indicators (QWI)
 - Quarterly separations and employment by state $\times$ 6-digit NAICS
 - 51 states + DC, years 2000–2023 (1,011 industries)
- **Quarterly analog to excess recurrence:** for each quarter $q \in \{1, 2, 3, 4\}$ and industry j ,

$$\text{Excess}(q)_j = \text{SepRate}_{j,q,t} - \frac{\text{SepRate}_{j,q-1,t} + \text{SepRate}_{j,q+1,t}}{2}$$

(quarters treated cyclically; average over years t)

- **Peak excess:** $\text{ExcessPeak}_j = \max_q \text{Excess}(q)_j$
- **Seasonal amplitude:** $\max_q - \min_q$ normalised to $[0, 1]$; does not assume any particular quarter is “the” seasonal quarter

National Seasonal Index: Top Industries

Table: Top 12 most seasonal 6-digit NAICS industries (national)

NAICS & description	ExcessPeak	n
111336 Fruit/tree nut combo farm.	0.333	18
722330 Mobile food services	0.316	18
111140 Wheat farming	0.287	20
111160 Rice farming	0.561	22
111199 Other grain farming	0.397	23
111219 Other veg. (ex. potato) farm.	0.459	15
111998 Other misc. crop farming	0.328	20
111120 Oilseed (ex. soybean) farm.	0.347	12
115113 Crop harvesting (machine)	0.398	10
512132 Drive-in motion picture theaters	0.362	21
532284 Recreational goods rental	0.485	22
111191 Oilseed/grain combo farm.	0.321	23

- n = years contributing to ExcessPeak estimate; a ≥ 5 -year minimum is enforced so thin, disclosure-suppressed cells can't masquerade as extreme seasonality
- Non-agriculture entries are intuitive once named: mobile food services (food trucks), drive-in theaters, recreational goods rental
- Seasonal index normalized to $[0,1]$ via 99th-pct winsorization
- Correlation with CP 1-digit rankings: $\rho = 0.829$

National Seasonal Index: Sector Summary

Table: Mean seasonal index by 2-digit sector (national)

Sector	Mean seasonal index	ExcessPeak (p.p.)
Agriculture, forestry, fishing	0.504	15.7
Arts, entertainment & recreation	0.461	14.3
Accommodation & food services	0.345	10.3
Educational services	0.272	8.4
Construction	0.194	5.9
Real estate	0.167	5.6
Warehousing & couriers	0.135	4.6
Other services	0.134	4.2
Health care & social assistance	0.067	2.2
Wholesale trade	0.063	1.8
Manufacturing (metals/machinery)	0.036	1.1

Education peaks in Q2 (end of school year), not Q4 — ExcessPeak is positive across all sectors. CP's 1-digit rankings confirmed at finer granularity.

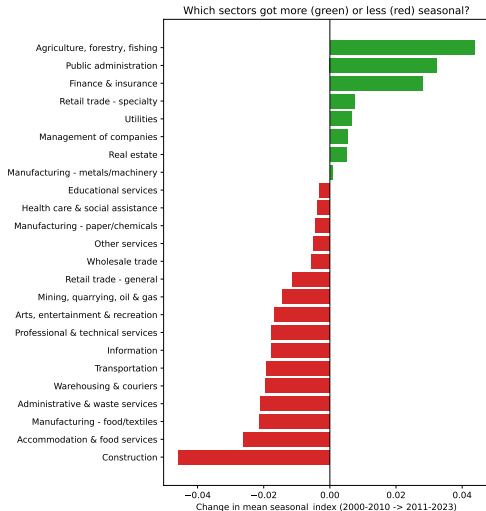
Robustness Check: Does COVID Distort the Index?

- The national index pools 2000–2023. Mass pandemic-era layoffs in industries already ranked highly seasonal (accommodation/food, arts & entertainment) could inflate “excess separation” for reasons that have nothing to do with normal annual seasonality
 - **Test:** recompute the index restricted to 2000–2019 only, compare to the full sample
 - **Result: rankings are robust.**
 - Pearson correlation (full vs. pre-COVID seasonal index): 0.997
 - Spearman rank correlation: 0.986
 - Only 6.3% of industries have a different peak quarter
 - Watch sectors (agriculture, construction, education, arts/entertainment, accommodation/food) all shift by ≤ 0.017
 - **One genuine mover:** a transportation industry (transit, NAICS 485111) whose index is roughly $5\times$ higher with COVID years included — plausibly a real ridership-collapse effect, not a data artifact
- Running this check is what surfaced a stale-data-threshold bug: three industries (incl. Tax Preparation Services, Cotton Ginning) had been wrongly ranked #1 nationally off just 2 years of COVID-distorted data, below the 5-year minimum — now fixed at the source

Seasonality Over Time: 2000–2010 vs. 2011–2023

- The national index pools all 24 years. Has seasonality actually been stable, or has it drifted over two decades?
 - **Test:** split the sample in two and recompute independently. The 5-year *n_excess_obs* minimum is what makes a clean two-way split possible — both halves (11 and 13 years) keep near-full coverage, so a three- or four-way split isn't advisable
 - **Result: correlated, with real drift.**
 - Pearson $r = 0.869$, Spearman $r = 0.795$ (both notably lower than the COVID check's $0.997/0.986$ — an 11-year gap between period midpoints is a much bigger ask than excluding 2 years)
 - Economy-wide mean seasonal index is essentially flat: $0.130 \rightarrow 0.126$
 - **But the flat average hides a real reshuffling** beneath it (next slide)
- 30.5% of industries have a different peak quarter, but this concentrates in weakly-seasonal industries (44% of the bottom quartile vs. 15% of the top) — ordinary noise, not genuine timing instability among strongly seasonal industries

Which Sectors Got More or Less Seasonal?



Agriculture: more seasonal

- +0.044, the largest sector increase (0.418 → 0.462)
- Plausibly consistent with growth in H-2A seasonal guest-worker visa usage over the 2010s (workers who leave every season, not year-round staff) — a hypothesis, not independently verified here

Construction: less seasonal

- -0.046, the largest sector decrease (0.216 → 0.170) — the single biggest shift either direction

Most others: mild decline

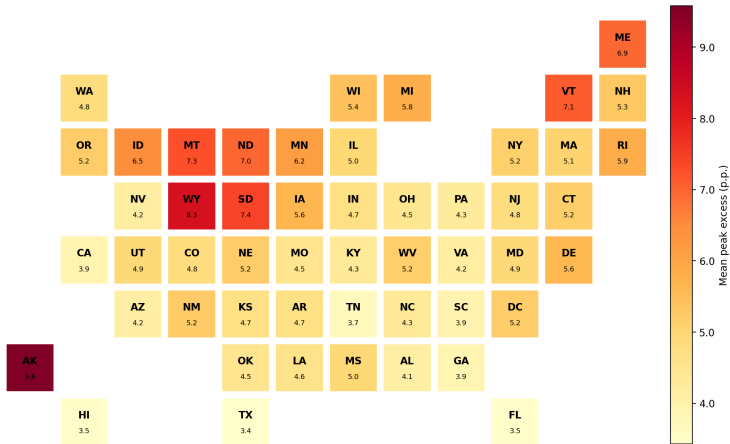
- Accommodation/food, transportation, professional services, arts/entertainment all -0.01 to -0.03

Geographic Variation: Overview

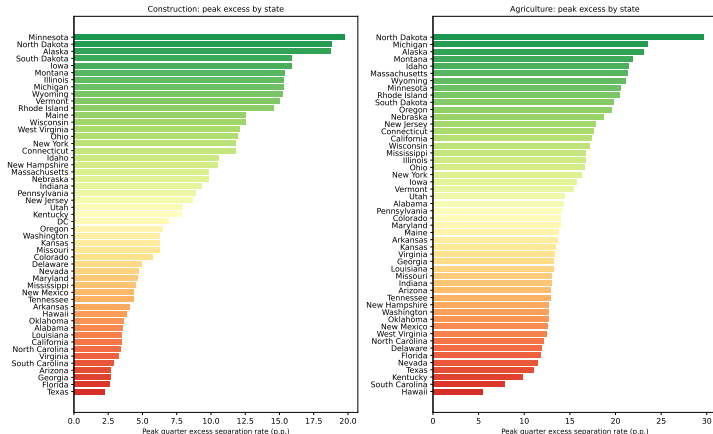
- The national index masks substantial state-level heterogeneity
- We compute ExcessPeak for each state $\times$ 6-digit NAICS cell (40,222 cells with ≥ 5 year-observations)
- **Main finding:** a $2.8\times$ gap between most and least seasonal states in mean peak excess (Alaska: 9.6 p.p. vs. Texas: 3.4 p.p.)
- Northern/Great Plains states consistently more seasonal across multiple sectors; Sun Belt states consistently less so
- **Most geographically variable sectors:**
 - ① Agriculture (cross-state SD = 14.5 p.p.)
 - ② Arts & entertainment (SD = 13.4 p.p.)
 - ③ Accommodation/food (SD = 10.8 p.p.)
 - ④ Construction (SD = 9.6 p.p.)
- **Least geographically variable:** healthcare, metals manufacturing, utilities — consistent low seasonality everywhere

State-Level Mean Seasonality

State-level seasonality
Mean peak excess separation rate across 6-digit NAICS industries



Construction and Agriculture by State



Construction: Minnesota (19.7 p.p.) vs. Florida (2.6 p.p.) — an $8\times$ gap driven by winter construction shutdowns in northern states. **Agriculture:** North Dakota (29.6 p.p.) now leads, ahead of Michigan (23.5 p.p.) and Alaska (23.1 p.p.); Montana and Idaho (previously #1/#2) remain in the top 5.

State-Specific Top and Bottom 1% Industries

Table: Within-state NAICS6 tails

Statistic	Top 1%	Bottom 1%
State-industry slots	413	413
Unique NAICS6 codes	133	252
Codes appearing once	57	158
Codes in 10+ states	6	0
Pairwise Jaccard	0.054	0.015
Also national same tail	11.4%	2.7%

Interpretation: the state-specific tails are related to the national index, but most state-tail industries are not simply the national top or bottom 1% repeated everywhere.

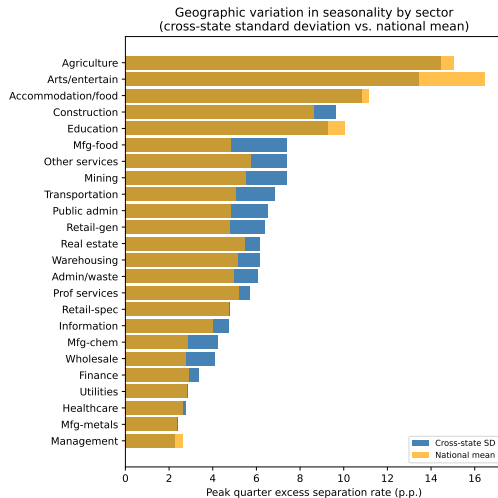
Recurring top-1% industries

- Nursery/tree production: 15 states
- Marinas: 14 states
- Specialty trade contractors: 12 states
- Golf courses/country clubs: 11 states
- Miscellaneous crop farming: 11 states

Bottom-1% pattern

- More diffuse than the top tail
- Concentrated in metals manufacturing, wholesale, chemicals, finance, and transportation

Geographic Variation: Sector Cross-State SD



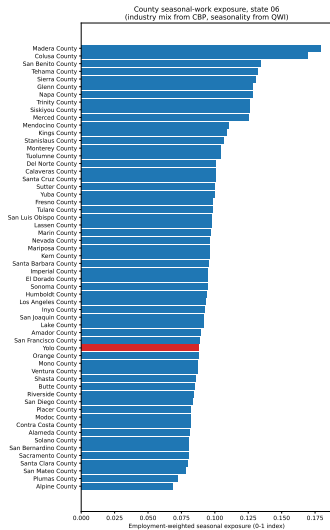
County-Level Seasonal Exposure: Methodology

- **Goal:** how seasonal is the average job in a given county, given the industries actually located there?

$$\text{Exposure}_c = \frac{\sum_j \text{EMP}_{c,j} \times \text{seasonal_index}_j}{\sum_j \text{EMP}_{c,j}}$$

- **Data:** County Business Patterns (CBP) establishment/employment counts by county $\times$ NAICS, merged onto the state $\times$ NAICS6 seasonal index
 - **Problem:** CBP suppresses most county $\times$ NAICS6 cells (disclosure avoidance) — a county's genuinely seasonal niche industries are often exactly the small, easily-identified cells that get suppressed
 - **Fix: three-tier fallback** for each industry's seasonal score: (1) state $\times$ NAICS6, (2) national NAICS6, (3) national NAICS4 (also recovers employment CBP suppresses at 6-digit but still reports at 4-digit)
- California pilot: **100% employment coverage** in all 58 counties — no county's score rests on a partial industry match

California County Ranking



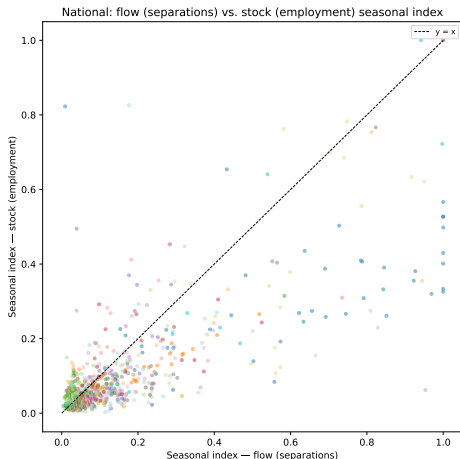
Case Study: Yolo County

- **Yolo County ranks 40th of 58** California counties (32nd percentile) — *less* exposed to seasonal work than roughly two-thirds of the state, despite being an agricultural county
 - **Why:** employment-weighting means large, stable service/logistics employers dominate the score:
 - Limited-service restaurants (3,222 jobs, index 0.10), general warehousing (3,087 jobs, index 0.06), couriers (1,731 jobs, index 0.14) — all low-seasonality but large employers
 - Crop harvesting scores the *maximum* 1.00 — but employs only 246 people locally
 - **Context:** Central Valley farm counties top the ranking (Madera 0.179, Colusa 0.170 — roughly 2× Yolo's 0.088); the rest of the Sacramento metro area (Sacramento, Solano, Placer) clusters near Yolo at the low-seasonality end
- A county can have extreme seasonal niches and still score low overall, if its broader economy has diversified around them

Flow vs. Stock: Motivation

- The index has always been built two ways in parallel, but the two were never formally compared:
 - **Flow** (`seasonal_index`): from separation *rates* — does a quarter see an excess of separations?
 - **Stock** (`seasonal_index_emp`): from employment *levels* normalized to their own annual mean — does headcount itself swell or shrink that quarter?
 - **Result: correlated, more so than first measured.**
 - National ($n = 997$): Pearson $r = 0.784$, Spearman $r = 0.660$
 - State $\times$ NAICS6 ($n = 39,446$): Pearson $r = 0.750$, Spearman $r = 0.661$
 - **Peak-quarter concordance is still low** — **$\sim 15\%$ nationally, $\sim 13\%$ by state**: even when both measures agree an industry is seasonal, they usually disagree on *which quarter* drives it
- Timing, not just amplitude, differs between “separations spike” and “headcount swings” — but the two are more tightly linked in overall magnitude than an earlier data-thinness bug made them look (see next slide)

Flow vs. Stock: Sector Heterogeneity



Construction: near-perfect agreement

- Flow/stock correlation $r = 0.97$ ($n = 31$):
winter layoffs = winter headcount drop

Agriculture: weaker, not decoupled

- $r = 0.64$ ($n = 55$), revised up sharply from an original $r \approx 0.21$ — that estimate was itself a data-thinness artifact (a few single-quarter cells dragging it down); the “rapid worker replacement” story is retracted

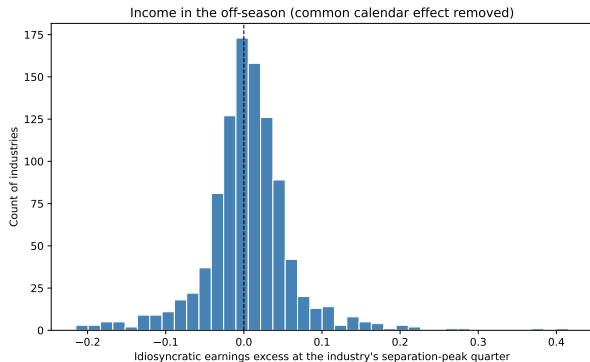
“Stock without flow,” updated

- Cotton Ginning (flow 0.01, stock 0.82) pairs with Political Organizations (flow 0.18, stock 0.83) — hiring surges (harvest; elections) without a matching separation spike. Tax Prep Services dropped out entirely (flow now correctly undefined)

A Third Lens: Does Income Dip in the Off-Season?

- Flow and stock both measure separations/headcount. A third QWI measure, EarnBeg (average monthly earnings), tests the paper's title question directly: **does income dip in an industry's own off-season?**
 - *Data note:* the originally-requested Payroll field returns null from the Census API for every cell tested (even huge industries where employment populates fine); EarnBeg is the working substitute
 - **A confound has to be fixed first:** 83% of *all* 1,011 industries — not just seasonal ones — have their raw earnings index peak in Q4. That's a broad, economy-wide year-end bonus/holiday-pay effect, not industry-specific timing
 - Left uncorrected, this swamps the signal: whether an industry's *own* separation-peak quarter happens to land in Q4 (everyone's bonus quarter) almost entirely determines whether it looks like income "dips" there
- **Fix:** compute the common cross-industry calendar effect (Q1 -0.031 , Q2 -0.005 , Q3 -0.048 , Q4 $+0.085$) and net it out before testing — the *idiosyncratic* index isolates industry-specific timing

Does Income Dip in the Off-Season? The Answer



Naive test: 51.9% "dip" — but almost entirely mechanical:

- Only 8.3% dip when the separation peak falls in Q4 (everyone's earnings peak)
- 93.6% "dip" when it falls in Q3 (rarely anyone's earnings peak)

De-confounded (credible) test:

- **43.9% dip**, mean excess $+0.0065$ — essentially flat, not clearly negative
- No strong evidence of industry-average income dipping in the off-season

Construction: the sharp counter-example

- Dip rate only 3.2%, mean excess $+0.044$ — earnings *higher*, plausibly overtime pay before winter layoffs

Doesn't contradict CP: their effect is the separating worker's own income loss (micro); this tests the industry-wide average (macro) — a different, complementary question.

Implications for Firm Classification

- The QWI index enables firm-level seasonal classification via NAICS lookup
- The top/bottom 1% analysis shows the exact NAICS6 tails differ sharply across states: top tail has 133 unique codes; bottom tail has 252
- **But:** for geographically sensitive sectors (construction, agriculture), a single national index may misclassify:
 - A Minnesota construction firm is highly seasonal
 - A Florida construction firm is nearly non-seasonal
- **Preferred approach for firm classification:**
 - ① Use the *state* $\times$ *NAICS* ExcessPeak as the lookup value when state of operation is known
 - ② Fall back to national index for industries with thin state-level coverage (low *n_excess_obs*)
 - ③ Direct estimation from UI wage records when sufficient separations are available ($n \geq 20$ per firm)
- **Which index?** Flow suits *turnover risk*; stock suits *headcount/staffing swings* — correlated ($r \approx 0.78$) but timing (peak quarter) still agrees only $\sim 15\%$ of the time
- **Caveat:** 6-digit cells are often suppressed in small states; state-level estimates for niche agriculture sub-industries may rest on very few year-observations

Summary

- 1 **Replication:** Python replication of CP (2025) excess recurrence matches all four published estimates within rounding
- 2 **National index:** QWI-based ExcessPeak at 6-digit NAICS correlates strongly with CP's 1-digit rankings ($\rho = 0.829$); robust to excluding 2020–2021 (Pearson 0.997). Economy-wide seasonality is roughly flat 2000–10 to 2011–23, but Agriculture rose (+0.044) and Construction fell (−0.046) beneath that average
- 3 **Geographic variation:** large state heterogeneity ($2.8\times$ gap, Alaska vs. Texas); a California county-level pilot applies the same logic down to counties — Yolo County ranks 40th of 58 CA counties, less seasonal than most despite its agricultural economy
- 4 **Flow vs. stock:** separations-based and employment-based seasonality are correlated ($r \approx 0.78$ national) but timing concordance is low ($\sim 15\%$); Construction near-identical ($r = 0.97$), Agriculture moderate ($r = 0.64$, revised up from an original 0.21 that turned out to be a data-thinness artifact)
- 5 **Income/earnings:** a third lens (EarnBeg) finds no strong evidence of industry-average income dipping in the off-season once a common Q4-bonus effect is removed; Construction again the outlier (earnings *higher*, not lower)
- 6 **Next steps:** national CBP rollout for a full county-level index; apply the state $\times$ NAICS index to UI wage records for firm-level classification